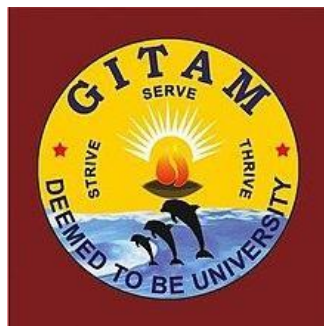


Application of Chemistry in Electronics (CHEM22411)

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Unit II: Advanced Electrochemical Energy Systems

Solid State Chemistry

8L

Introduction, classification and properties of solids. Crystallographic systems, types of lattices, Brag's equation, Born-Haber cycle and cohesive energy. Ionic and liquid crystals - properties and applications

Introduction, classification and properties of solids.

Solids are a state of matter characterized by fixed shape and volume, high density, and incompressibility due to strong intermolecular forces and tightly packed constituent particles. They can be classified as either [crystalline solids](#) with ordered, repeating arrangements of particles and sharp melting points, or [amorphous solids](#) with disordered arrangements and softening over a range of temperatures.

Introduction to Solids:

Solids are a fundamental state of matter where particles are held together by strong intermolecular forces, leading to their characteristic rigidity and resistance to changes in shape and volume. They cannot flow like liquids or gases and are virtually incompressible due to the negligible space between their constituent particles.

Classification of Solids:

Solids are primarily classified into two main categories based on the arrangement of their constituent particles:

1. Crystalline Solids: These solids have a well-defined, long-range, ordered arrangement of particles, forming a regular, repeating pattern known as a crystal lattice. Examples include sodium chloride (NaCl), quartz, and metals. They exhibit sharp and characteristic melting points.

2. Amorphous Solids: These solids lack a definite, long-range ordered arrangement of particles and are characterized by a disordered structure. Examples include glass, rubber, and plastics. They are sometimes referred to as "pseudo solids" or "supercooled liquids" and soften gradually over a range of temperatures instead of having a sharp melting point.

Properties of Solids:

Solids possess several defining properties stemming from the strong intermolecular forces and close packing of their particles:

- Rigidity: They are rigid structures and resist changes in their shape when external forces are applied.
- Definite Shape and Volume: Solids have a fixed mass, volume, and form.
- High Density: Due to the close packing of their constituent particles, solids generally have high densities compared to liquids and gases.
- Incompressibility: The negligible intermolecular distances make solids virtually incompressible.
- Low Compressibility: Similar to incompressibility, solids have a very low tendency to change their volume under pressure.
- High Melting Points: The strong forces holding the particles together require significant energy to overcome, leading to high melting points.
- Oscillation of Particles: The particles in a solid are not stationary but can only oscillate about their fixed mean positions.
- Negligible Diffusion: The rate of diffusion of one solid into another is very low and takes a significant amount of time.
- Electrical Properties: Solids can be classified based on their electrical conductivity as conductors (e.g., metals), insulators, or semiconductors.

Types of Crystalline Solids

On the basis of the nature of intermolecular forces or [chemical bonding](#), crystalline solids are further classified into four categories. They are as follows:

- Molecular solids
- Ionic solids
- Metallic solids
- Covalent solids

Molecular Solids

In molecular solids, the constituent particles are molecules. They are further divided into three categories:

1. Non-polar Molecular Solids

These solids are formed from molecules or atoms that share a [non-polar covalent bond](#). The atoms or molecules are held by weak dispersion force or by London forces.

- The physical nature of non-polar solids is soft.
- They don't conduct electricity, so they are insulators.
- They have a very low melting point.
- Examples: H_2 , Cl_2 , I_2 etc.

2. Polar Molecular Solids

- These solids are held together by [polar covalent bonds](#), and the atoms/molecules are bonded by relatively stronger dipole-dipole interactions.
- The physical nature is soft, and most of these are gases or liquids at room temperature.
- They do not conduct electricity, and they have a higher melting point than non-polar molecular solids.
- Examples: HCl , SO_2 , NH_3 , etc.

3. Hydrogen-bonded Molecular Solids

The solids contain polar covalent bonds with Hydrogen, Fluorine, Oxygen and Nitrogen atoms. In these solids, molecules are held together via strong [hydrogen bonding](#).

- The physical nature of such solids is hard.
- They do not conduct electricity.
- The physical state of these solids is volatile liquids or soft solids under room temperature.
- They have a low melting point.
- Example: H₂O (Ice).

Ionic Solids

In ionic solids, the constituent particles are ions. These are formed by the arrangement of cations and anions by strong Coulombic forces.

These are hard and brittle in nature.

Ionic solids act as an insulator in a solid-state but are conductors in a molten and aqueous state.

They have a high melting point.

Examples: NaCl, MgO, ZnS, CaF₂ etc.

Metallic Solids

Positive metal ions in a sea of delocalised electrons. These electrons are evenly spread out throughout the crystal.

Due to the presence of free and mobile electrons, they are responsible for high electrical and thermal conductivity.

They are conductors in both solid and molten states.

The physical nature of these solids is hard, but they are malleable and ductile.

They have high melting point than ionic solids.

Examples: Fe, Cu, Ag, Mg, etc.

Covalent or Network Solids

A wide range of crystalline solids of non-metal form [covalent bonds](#) between adjacent atoms throughout the crystal and form giant molecules or large molecules.

- These solids are hard, like diamond and soft, like graphite which are isotopes of carbon.
- They are insulators, as in the case of a diamond, but in the case of graphite, due to free electrons, they conduct electricity and act as a conductor.

Amorphous Solid State

Amorphous solid-state comprises those solids which have the property of rigidity and incompressibility but to a certain extent. They do not have a definite geometrical form or long range of order. Examples include glass, rubber, plastic, etc.

Properties of Amorphous Solids

- Amorphous solids are gradually softened over a range of temperatures, and they can be moulded into different shapes on heating.
- Amorphous solids are pseudo-solids or supercooled liquids, which means they have a tendency to flow very slowly. If you observe the glass pans, which are fixed to windows of old buildings, they are found to be slightly thicker from the bottom than at the top.
- Amorphous solids have irregular shapes. i.e., their constituent particles do not have definite geometry of arrangements.
- When amorphous solids are cut with a sharp edge tool, they form pieces with irregular surfaces.
- Amorphous solids do not have definite heat of fusion due to the irregular arrangement of the particles.
- Amorphous solids are isotropic in nature, which means the value of any physical property would be the same along any direction because of the irregular arrangement of particles.

Amorphous Solids Uses

Amorphous silicon, which is one of the best photovoltaic materials, converts sunlight into electricity.

1. Born–Haber Cycle

The **Born–Haber cycle** is a thermochemical cycle used mainly for calculating the **lattice energy** of an ionic solid, using Hess's law.

It decomposes the formation of an ionic compound from its elements into several hypothetical steps whose enthalpy changes are known or measurable.

For example, for **NaCl(s)**:

Steps:

1.Sublimation of solid Na $\rightarrow$ Na(g)

ΔH_{sub} (endothermic)

2.Ionization of Na(g) $\rightarrow$ Na⁺(g) + e⁻

I_1 (ionization energy, endothermic)

3.Dissociation of Cl₂(g) $\rightarrow$ 2 Cl(g)

$\frac{1}{2} \Delta H_{\text{diss}}$ (endothermic)

4.Electron affinity of Cl(g) + e⁻ $\rightarrow$ Cl⁻(g)

EA (exothermic)

5.Lattice formation from gaseous ions: Na⁺(g) + Cl⁻(g) $\rightarrow$ NaCl(s)

Lattice energy (U, exothermic)

Overall equation:

$$\Delta H_{\text{f}}^\circ = \Delta H_{\text{sub}} + I_1 + \frac{1}{2} \Delta H_{\text{diss}} + \text{EA} + U$$

Rearranging lets you solve for **U** if $\Delta H_{\text{f}}^\circ$ and the other quantities are known.

2. Cohesive Energy

Cohesive energy is the energy required to completely separate a solid into isolated free atoms at rest, at **0 K**. It measures the **binding strength** of atoms in a crystal (metallic, covalent, ionic, or molecular).

For an **ionic crystal**, cohesive energy includes:

- **Lattice energy** (electrostatic contribution)
- Other binding contributions (covalent, polarization, etc.)

Mathematically:

$$E_{\text{coh}} = (\text{Sum of energies of isolated atoms}) - (\text{Energy of the crystal})$$
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For ionic solids, cohesive energy is **numerically close to** lattice energy, but they're **not identical** because cohesive energy is usually defined at 0 K (purely potential + zero-point vibration), while lattice energy is defined for gaseous ions combining into a crystal at 298 K.

3. Connection Between Born–Haber and Cohesive Energy

In **ionic solids**, the Born–Haber cycle gives the **lattice energy** U_{L} , which is a major part of the **cohesive energy**.

Cohesive energy can be obtained from **lattice energy** plus the additional energies required to produce the gaseous atoms (sublimation, dissociation, ionization, electron affinity steps in reverse).

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EA (exothermic)

5.Lattice formation from gaseous ions: Na⁺(g) + Cl⁻(g) $\rightarrow$ NaCl(s)

Lattice energy (U, exothermic)

The **Born Haber process**, more commonly referred to as the **Born Haber cycle**, is a method that allows us to observe and analyse energies in a reaction. It mainly helps in describing the formation of ionic compounds from different elements. The methodology further enables us to understand the overall reaction process through a series of steps.

The Born Haber cycle was introduced in the year 1919 by German scientists Fritz Haber and Max Born. The Born Haber cycle is mainly used to calculate the [lattice energy](#). It also involves several steps or processes, such as electron affinity, ionization energy, sublimation energy, the heat of formation and dissociation energy.

Considerations

The reaction of electropositive metals with electronegative nonmetals produces ionic solids. Alkali and alkaline earth metals react with chalcogen or halogen family elements to form compounds, which are crystalline ionic solids. Ionic compounds being stabilized by the electrostatic force of attraction between positive and negative charges are expected to have similar physical properties.

Lattice energy is the energy that keeps together the cations and anions of the compound in fixed positions in a crystalline solid state. Lattice energy can be defined as either energy released when gaseous ions form one mole of a solid ionic compound or as the energy required to convert one mole ionic solid into its gaseous ions. There is no way to measure this lattice energy experimentally. Hess's law of heat summation is the only indirect way of estimating the lattice energy.

Application of [Hess's law of heat summation](#) to the formation of solid ionic compounds involves enthalpy of all processes that are necessary for the formation of the solid ionic compound from the elemental state of the constituent atoms in a form cycle such that the total energy on summation is zero.

What Is the Born Haber Cycle?

The Born Haber cycle is a cycle of enthalpy change of process that leads to the formation of a solid crystalline ionic compound from the elemental atoms in their standard state and of the enthalpy of formation of the solid compound such that the net enthalpy becomes zero

Born Haber Cycle Examples

Example 1: Born Haber cycle of sodium chloride NaCl (or any AB-type Mono-valent ionic solid).

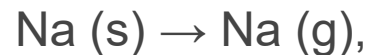
The heat of the formation of sodium chloride (ΔH_f^0) from the sodium metal and chlorine gas can be experimentally measured.



The formation of ionic solid sodium chloride from solid sodium metal and gaseous chlorine is not a single-step process but goes through several processes. Heat changes of all the processes except the lattice energy can be experimentally measured.

The processes or steps in the formation of sodium chloride are as follows:

1. Solid sodium atom sublimates to gaseous atom by absorbing heat energy (ΔH_{sub}).



Sublimation energy $\Delta H_{\text{sub}} = + 107\text{kJ/mol}$

2. Gaseous sodium atom absorbs the ionization energy to release one electron and forms a gaseous sodium ion.



Ionization energy $\Delta H_{\text{IE}} = +502\text{kJ/mol}$

3. Diatomic gaseous chlorine breaks into two individual atoms by absorbing bond energy, such that each chlorine atom absorbs half of the bond energy of the chlorine molecule.



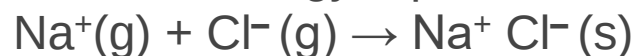
Bond dissociation energy of chlorine = $(1/2) \Delta H_{\text{diss}} = (1/2) 242 = +121\text{kJ/mol}$

4. Chlorine atom accepts an electron to form a chloride ion and releases energy equivalent to electron affinity.



Electron affinity = $\Delta H_{\text{EA}} = -355\text{kJ/mol}$

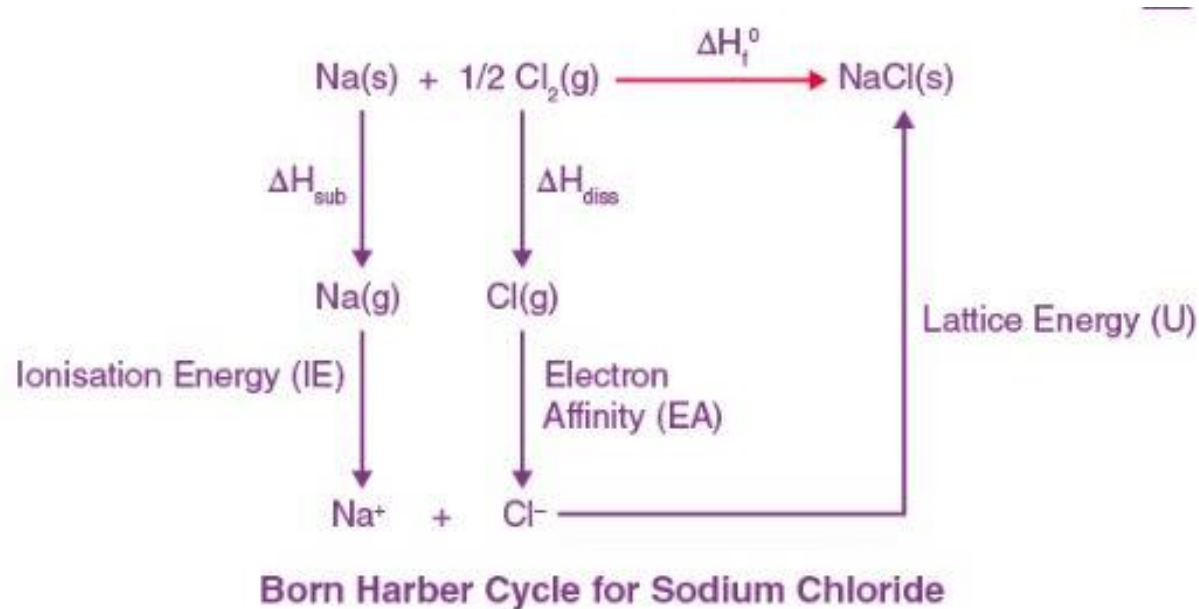
5. Gaseous sodium ion and gaseous chloride ion combine to form a solid sodium chloride molecule and releases energy equivalent to lattice energy.



Lattice energy = $\Delta H_{\text{LE}} = U = ?$

The summation of enthalpy of all the processes (from step 1 to step 5) give the net enthalpy of the formation of solid crystalline sodium chloride from sodium and chlorine in their standard conditions of solid and gas, respectively. This should be equal to the experimentally measured enthalpy of the formation of solid sodium chloride.

The enthalpies are represented as a cycle in the figure.



So, $\Delta H_f^0 = \Delta H_{\text{sub}} + \Delta H_{\text{IE}} + (1/2) \Delta H_{\text{dis}} + \Delta H_{\text{EA}} + U$ or $\Delta H_f^0 - (\Delta H_{\text{sub}} + \Delta H_{\text{IE}} + (1/2) \Delta H_{\text{dis}} + \Delta H_{\text{EA}} + U) = 0$

$$411 + 107 + 502 + 121 - 355 + U = 0$$

Here, except lattice energy, all other enthalpies can be experimentally measured.

The lattice energy of the sodium chloride solid = $U = \Delta H_f^0 - (\Delta H_{\text{sub}} + \Delta H_{\text{IE}} + \frac{1}{2} \Delta H_{\text{dis}} + \Delta H_{\text{EA}})$.

$$= -411 - 107 - 502 - 121 + 355$$

$$= -786 \text{ kJ/mol}$$

2. Cohesive Energy

Cohesive energy is the energy required to completely separate a solid into isolated free atoms at rest, at **0 K**. It measures the **binding strength** of atoms in a crystal (metallic, covalent, ionic, or molecular).

For an **ionic crystal**, cohesive energy includes:

- **Lattice energy** (electrostatic contribution)
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Mathematically:

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3. Connection Between Born–Haber and Cohesive Energy

- In **ionic solids**, the Born–Haber cycle gives the **lattice energy** U_{UUU} , which is a major part of the **cohesive energy**.
- Cohesive energy can be obtained from **lattice energy** plus the additional energies required to produce the gaseous atoms (sublimation, dissociation, ionization, electron affinity steps in reverse).

✓ **Key differences:**

Aspect	Born–Haber Cycle	Cohesive Energy
Purpose	Calculate lattice energy	Measure total bonding strength in solid
Method	Hess’s law thermochemical cycle	Energy difference between solid & free atoms
Scope	Mostly ionic solids	Any solid type
Units	kJ mol^{-1}	kJ mol^{-1}